Лек9: БЧХ-кода и РС-кода

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$d \le r+1 = n-k+1$

1 –

2 – код с проверкой четности

3 – код Хэмминга

4 – расширенный код Хэмминга

$(7,4)$ $\quad 2^3=8$

$$H = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$GF(2)$ – $1+x+x^3 = p(x)$

$GF(2^3)$

1

x

$1+x$

x^2

x^2+1

x^2+x

x^2+x+1

α^0	1
α	x
α^2	x^2
α^3	$1+x$
α^4	$x+x^2$
α^5	$1+x+x^2$
α^6	$x+x^2$

$H = (1 \ \alpha \ \alpha^2 \ \alpha^3 \ \alpha^4 \ \alpha^5 \ \alpha^6)$

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

$(15, 12)$ $\quad n = n - k$ $\qquad GF(2) \to GF(2^4)$

$H = (1 \ \alpha \ \alpha^2 \ldots \alpha^{14})$ $\qquad p = 1 + x + x^4$

$c(x) + e(x) = b(x)$

$$s = \sum_{i=0}^{n-1} b_i \alpha^i = \sum_{i=0}^{n-1} (c_i + e_i)\alpha^i = \sum_{i=0}^{n-1} e_i \alpha^i = e(\alpha)$$

В позиции i - ошибка $e(x) = x^i$, $s = \alpha^i$

$e(x) = x^i + x^j$, $i \neq j$

$s = \alpha^i + \alpha^j$

$$\begin{cases} s_1 = \alpha^i + \alpha^j \\ s_2 = f(\alpha^i) + f(\alpha^j) \end{cases} \qquad H = \begin{pmatrix} 1 & \alpha & \alpha^2 & \ldots & \alpha^{14} \\ \beta_1 & \beta_2 & \beta_3 & \ldots & \beta_{15} \end{pmatrix}$$

$\beta_1 \in GF(2^m)$

$f(\alpha) = c\alpha$ $\qquad GF(2)$

$f(\alpha) = \alpha^2$ $\qquad b(x^2) = b^2(x)$

$1^2 = 1$, $0^2 = 0$

$$H = \begin{pmatrix} 1 & \alpha & \alpha^2 & \alpha^3 & \alpha^4 & \alpha^5 & \alpha^6 & \alpha^7 & \alpha^8 & \alpha^9 & \alpha^{10} & \alpha^{11} & \alpha^{12} & \alpha^{13} & \alpha^{14} \\ 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} & 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} & 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} \end{pmatrix}$$

i, j - позиции ошибок $\quad x = \alpha^i$, $y = \alpha^j$

$$\begin{cases} x + y = s_1 \\ x^3 + y^3 = s_2 \end{cases}$$

при одной или двух ошибка

$s_1 \neq 0$

$$\frac{x^3 + y^3}{x + y} = \frac{s_2}{s_1} \qquad (x^3 + y^3) = (x + y)(x^2 + y^2 + xy)$$

$$(a+b)^2 = a^2 + 2ab + b^2 = a^2 + b^2$$

$$x^2 + y^2 + xy = \frac{S_2}{S_1}$$

$$x + y = S_1 \qquad (x+y)^2 = S_1^2 \qquad x^2 + y^2 = S_1^2$$

$$S_1^2 + xy = \frac{S_2}{S_1}$$

$$xy = \frac{S_2}{S_1} - S_1^2$$

Т Виета

$$\begin{cases} xy = \frac{S_2}{S_1} - S_1^2 \\ x + y = S_1 \end{cases} \qquad \begin{cases} z^2 + \delta_1 z + \delta_2 = 0 \\ \delta_1 = S_1, \ \delta_2 = \frac{S_2}{S_1} - S_1^2 \end{cases}$$

$GF(2) \to GF(2^4) \qquad p(x) = 1 + x + x^4$

Степени α	Полином	двоич. пол.
1	x	0010
2	x^2	0100
3	x^3	1000
4	$1+x$	0011
5	$x+x^2$	0110
6	x^2+x^3	1100
7	$1+x+x^3$	1011
8	$1+x^2$	0101
9	$x+x^3$	1010
10	$1+x+x^2$	0111
11	$x+x^2+x^3$	1110
12	$1+x+x^2+x^3$	1111
13	$1+x^2+x^3$	1101
14	$1+x^3$	1001

$$H = \begin{pmatrix} 1 & \alpha & \alpha^2 & \alpha^3 & \alpha^4 & \alpha^5 & \alpha^6 & \alpha^7 & \alpha^8 & \alpha^9 & \alpha^{10} & \alpha^{11} & \alpha^{12} & \alpha^{13} & \alpha^{14} \\ 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} & 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} & 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} \end{pmatrix}$$

$$H = \begin{array}{l} 000100 \\ 001001 \\ 010011 \quad \rightarrow \\ \underline{100010} \\ 011110 \\ 001010 \quad \rightarrow \\ 000110 \\ 100011 \end{array}$$

$$\begin{array}{ccccccccccccccc} 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \\ \rightarrow 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 1 \end{array}$$

$S_1 = \alpha^2 + \alpha^3 = \alpha^{11}$, $S_2 = \alpha^5 + \alpha^{12} = \alpha^4$

$$\begin{cases} xy = \frac{S_2}{S_1} - S_1^2 \\ x + y = S_1 \end{cases} \qquad \frac{\alpha^4}{\alpha^{11}} + (\alpha^{11})^2 = \frac{1}{\alpha^7} + \alpha^7 = \alpha^8 + \alpha^7 = \alpha^{11}$$

$$= \frac{1 + \alpha^{14}}{\alpha^7} = \frac{\alpha^3}{\alpha^7} = \alpha^{-4}$$

$\delta^2 = \alpha^{11}$, $\delta^1 = \alpha^{11}$

$z^2 + \alpha^{11} \cdot z + \alpha^{11} = 0$ $\qquad \alpha^2, \alpha^9$

$(\alpha^2)^2 + \alpha^{11} \cdot \alpha^2 + \alpha^{11} = \alpha^4 + \alpha^{13} + \alpha^{11}$

$= 1 + x + 1 + x^2 + x^3 + x + x^2 + x^3 = 0$

$(\alpha^9)^2 + \alpha^{11} \cdot \alpha^9 + \alpha^{11} = \alpha^3 + \alpha^5 + \alpha^{11}$

$= x^3 + x + x^2 + x + x^2 + x^3 = 0$